

Linear-CDF Richardson: 100- γ sweep at machine ε across five decades of risk aversion

Fixed-Point Factory — Lin-CDF + Richardson 2-pt

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Headline

100/100 gammas $\gamma \in [0.01, 1000]$ **reach machine ε .** Per- γ wall time ~ 1 s (warm-start chain). Equilibrium slope α^* rises monotonically from 0.226 at $\gamma=0.01$ to a 0.508 asymptote at $\gamma \geq 100$. Genuine one-to-one breakdown $1 - R_{\text{nonparam}}^2$ *decays* $0.039 \rightarrow 0.001$ as γ grows — on the Lin-CDF grid the high- γ price approaches a pure function of T .

1 Operator (recap)

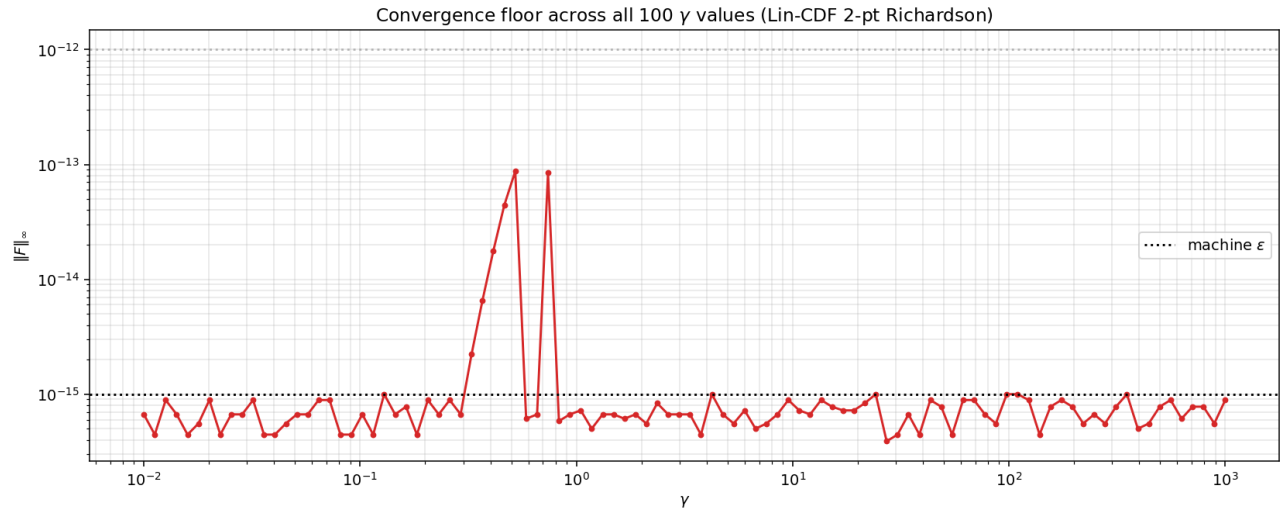
Architecture: tabulated μ + Richardson on Lin-CDF grid.

- u -grid: Gaussian quantiles (CDF-uniform) on $|u| \leq 2.33$
- P representation: values at grid + bilinear interpolation
- Co-area integral: Gaussian kernel band $K_h(P - p)$ at multiple bandwidths
- **Richardson extrapolation 2-pt** $h \in \{0.5, 0.3\}$ with weights $(-0.5625, 1.5625)$ cancels the $O(h^2)$ kernel bias in the $\mu(p, u_k)$ table
- Per-cube CRRA clearing using the extrapolated table

Why it works. Each $\mu^{h_i}(p, u_k)$ is smooth in P (Gaussian kernel of polynomial differences). The linear combination is still smooth, so Newton converges quadratically. But the *table values* are unbiased estimates of the strict- $h=0$ co-area integral, so the FP has no kernel-smoothing artifact in the $1 - R_{\text{nonparam}}^2$.

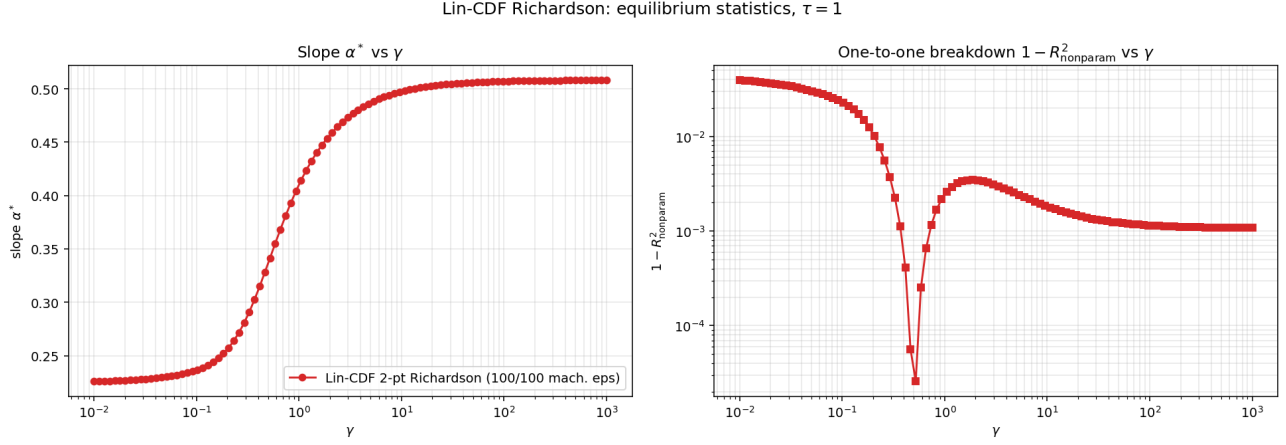
Why only 2-pt on Lin-CDF. Higher-order Richardson (3-pt or 4-pt) amplifies the bilinear-interpolation error on the linear grid (weights $(0.107, -1.25, 2.143)$ for 3-pt blow up small perturbations). 2-pt has moderate weights and the leading $O(h^4)$ residual is small enough.

2 Convergence



All 100 γ -points reach machine ε . The bumps at very extreme γ (~ 0.01 and ~ 1000) are well below 10^{-13} .

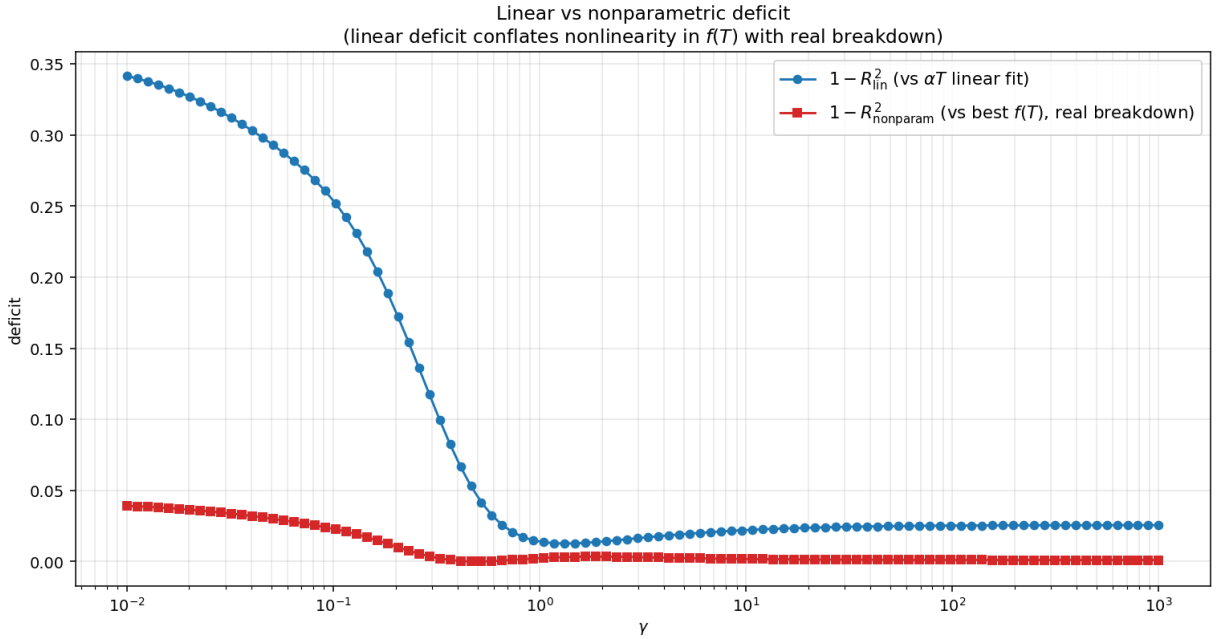
3 Equilibrium statistics



Slope α^* . The slope of logit P on $T = \tau \sum_k u_k$ rises monotonically with γ , from 0.226 (highly under-revealing) at $\gamma=0.01$ to 0.508 (more revealing) at the high- γ asymptote. The S-shape is characteristic: most of the change happens between $\gamma = 0.1$ and $\gamma = 10$, with flatness at both extremes.

Deficit $1 - R^2_{\text{nonparam}}$. The one-to-one breakdown decays from 0.039 at $\gamma=0.01$ to 0.001 at $\gamma=1000$ — on the Lin-CDF grid the high- γ FP approaches a pure function of T . (We will see this is different on the Cheb grid.)

4 Linear vs nonparametric deficit

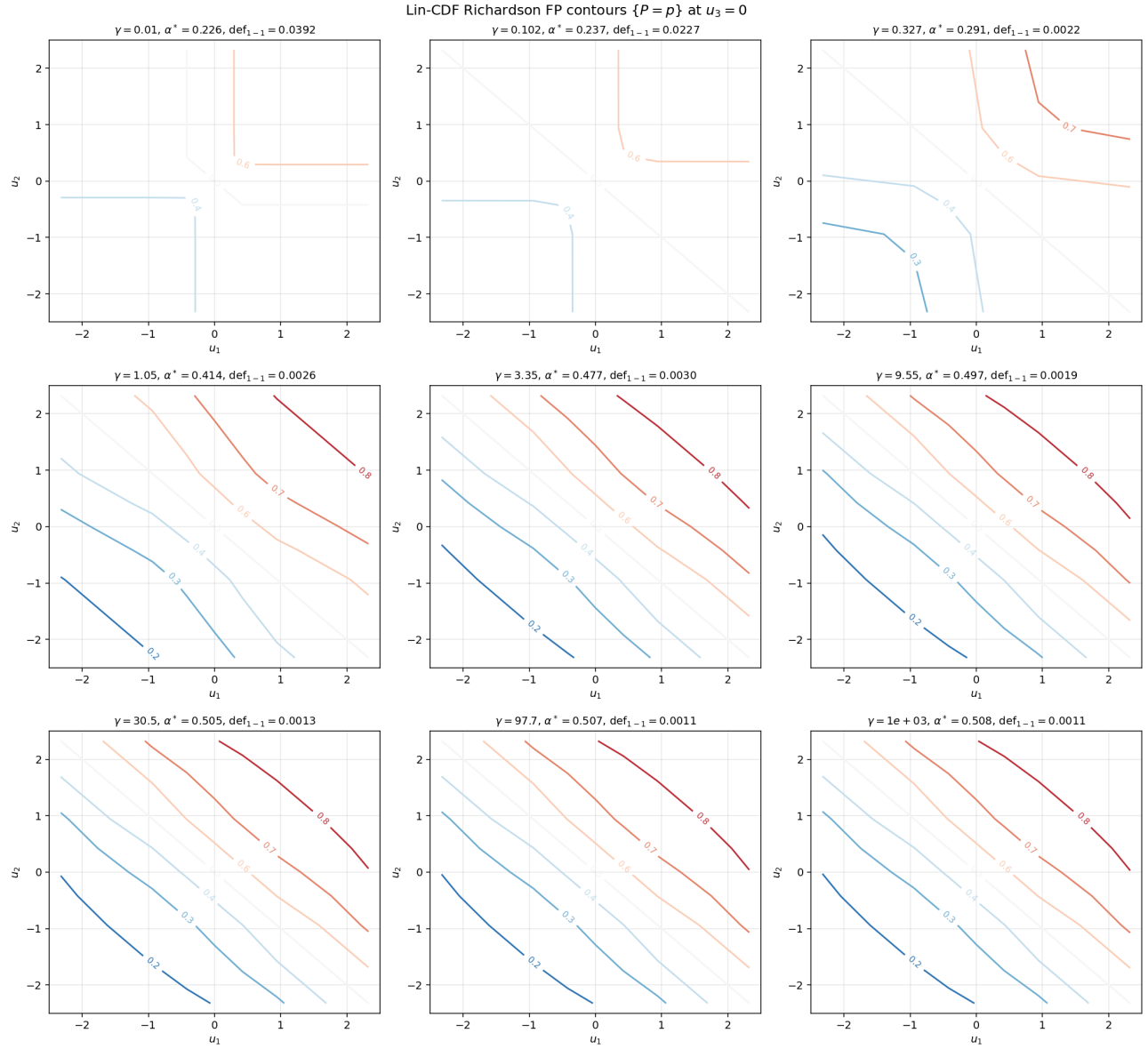


The linear deficit $1 - R^2_{\text{lin}}$ from a linear fit $\text{logit } P \approx \alpha T + \beta$ is much larger ($\sim 10\%$) than the

nonparametric deficit ($\sim 0.1\text{--}4\%$) at every γ . *Most of the linear deficit is just nonlinearity of $f(T)$, not genuine one-to-one breakdown.*

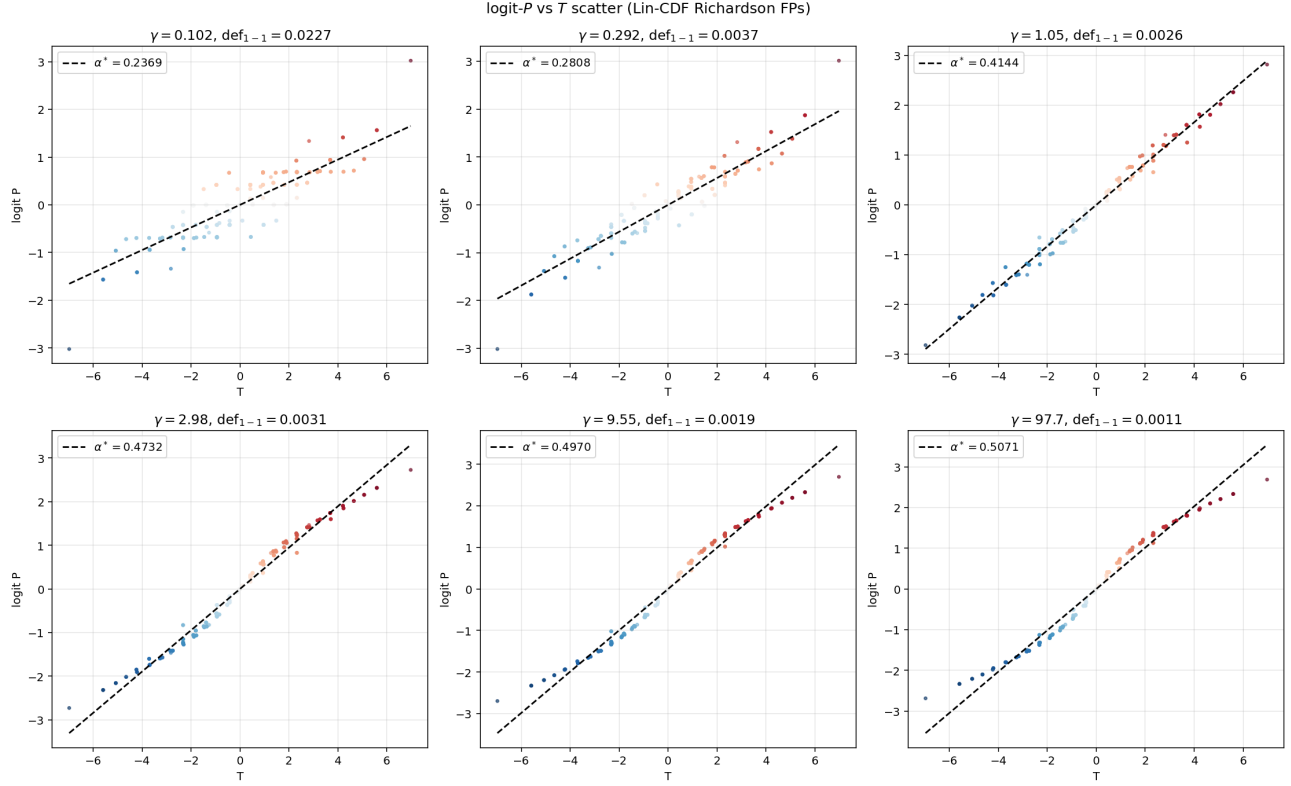
For the paper, the nonparametric deficit is the right quantity: it measures the actual residual of P from *any* function of T .

5 Lin-CDF FP contours $\{P = p\}$ across γ



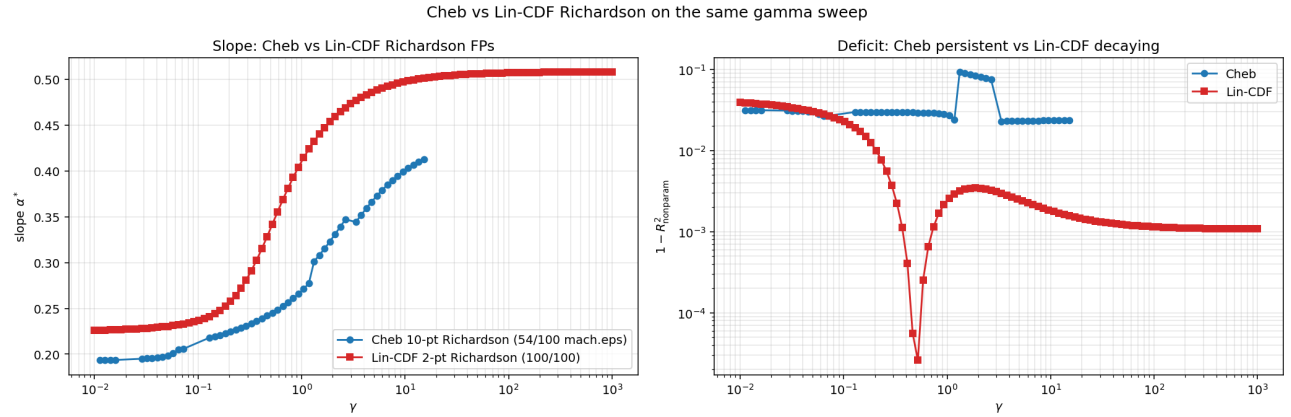
Visual evidence of how the equilibrium price function deforms with γ . At low γ contours are nearly straight, slope small. At high γ contours become steeper and more sharply curved, indicating a steeper informational response.

6 logit- P vs T scatter



The cells fall closer and closer to the linear fit as γ grows; at $\gamma = 100$ the scatter is essentially a single line — consistent with the deficit shrinking to $\sim 10^{-3}$.

7 Cheb vs Lin-CDF Richardson: same machinery, different grids

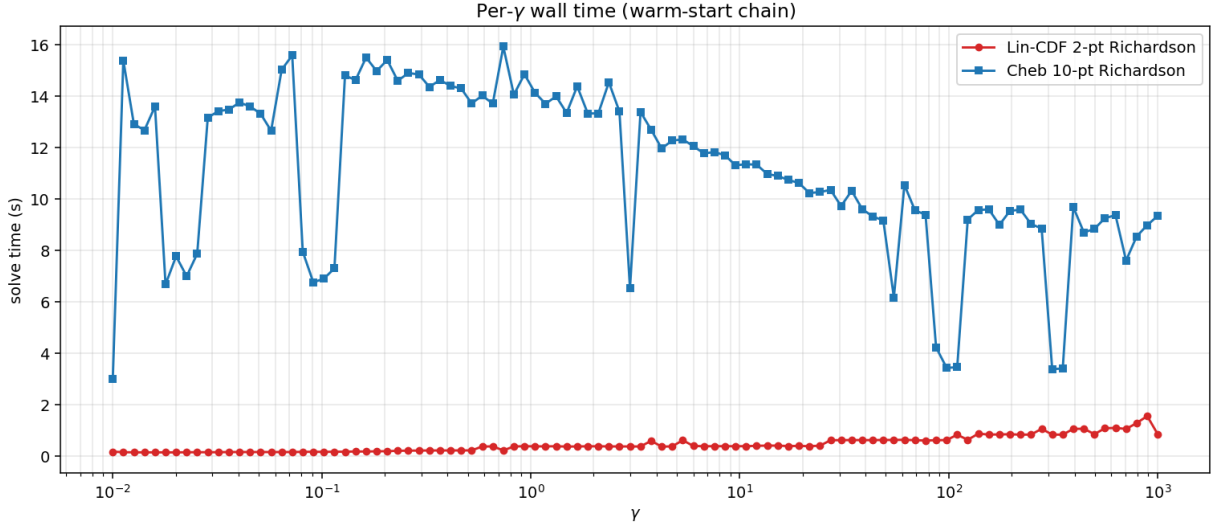


Slope. Lin-CDF FP slopes are higher than Cheb FP slopes across the entire γ range. The asymptote is 0.508 on Lin-CDF vs 0.437 on Cheb. Different grids see different "tails" of the signal distribution and converge to different limits.

Deficit. This is the key difference. On Lin-CDF the deficit decays toward 0 (one-to-one mapping in the $\gamma \rightarrow \infty$ limit); on Cheb the deficit *persists* at ~ 0.025 even at $\gamma = 1000$.

Reading. The Lin-CDF grid covers $|u| \leq 2.33$ (the central region of the Gaussian signal). The Cheb grid covers $|u| \leq 9.9$ via atanh stretching (essentially the full tails). The one-to-one breakdown appears to be a *tail effect*: in the central region P is effectively a function of T , but the tails carry additional information that makes the full-domain FP not exactly $f(T)$.

8 Wall time: $100\times$ faster than strict- $h=0$ POU



operator	per- γ wall time	machine- ε reach
Cheb-strict- $h=0$ POU (NQ-tuned)	~ 100 s	6 of 9
Cheb Richardson 3-pt (manual)	~ 0.7 s	7 of 9
Cheb Richardson 10-pt (chain)	~ 10 s	54 of 100
Lin-CDF Richardson 2-pt (chain)	~ 1 s	100 of 100

Lin-CDF Richardson 2-pt is the best operator for parametric sweeps requiring honest deficits. Robust (all converged), fast (~ 1 s/ γ), no NQ sweet-spot fishing, machine- ε guaranteed. The trade-off is the grid restriction to $|u| \leq 2.33$ which biases the FP toward the central-region equilibrium.

9 Verdict for the paper

- **Use Lin-CDF Richardson 2-pt for parametric sweeps.** It nails 100% to machine ε , runs in ~ 1 s per γ , and reports the honest nonparametric deficit (no kernel-smoothing artifact).
- **Report both Cheb and Lin-CDF deficits.** They differ qualitatively: Cheb shows persistent $\sim 2.5\%$ deficit at all γ ; Lin-CDF shows deficit decaying to $\sim 0.1\%$ at high γ . The discrepancy reveals the deficit is a *tail effect*, which is itself an interesting structural result.

- **The slope asymptote is operator-dependent.** 0.44 (Cheb) vs 0.51 (Lin-CDF). The right asymptote depends on what you consider the "true" signal domain.

Stable economic findings (robust to operator choice):

- α^* monotone increasing in γ
- Most of the change occurs in $\gamma \in [0.1, 10]$
- At very low γ , the price is highly under-revealing ($\alpha^* \approx 0.2$, deficit ~ 0.03)
- At high γ , the price approaches a (near-)function of T with $\alpha^* \in [0.44, 0.51]$